

AUTOMORPHIC FORMS

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1. MODULAR FORMS

Definition 1.0.1. (♠ TODO: define congruence subgroups for more general arithmetic groups and describe how these are instances of those) Let N be a positive integer.

- The *principal congruence subgroup of $\mathrm{SL}_2(\mathbb{Z})$ of level N* is the group $\Gamma(N)$ defined by

$$\Gamma(N) := \{\gamma \in \mathrm{SL}_2(\mathbb{Z}) : \gamma \equiv I_2 \pmod{N}\}.$$

- The subgroup $\Gamma_1(N)$ of $\mathrm{SL}_2(\mathbb{Z})$ is defined by

$$\Gamma_1(N) := \left\{ \gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{SL}_2(\mathbb{Z}) : a \equiv d \equiv 1 \pmod{N}, c \equiv 0 \pmod{N} \right\}.$$

- The subgroup $\Gamma_0(N)$ of $\mathrm{SL}_2(\mathbb{Z})$ is defined by

$$\Gamma_0(N) := \left\{ \gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{SL}_2(\mathbb{Z}) : c \equiv 0 \pmod{N} \right\}.$$

- A *congruence subgroup of $\mathrm{SL}_2(\mathbb{Z})$* is any subgroup $\Gamma \leq \mathrm{SL}_2(\mathbb{Z})$ that contains $\Gamma(N)$ for some $N \geq 1$.

Note that

$$\Gamma(N) \subset \Gamma_1(N) \subset \Gamma_0(N) \subset \mathrm{SL}_2(\mathbb{Z}).$$

Moreover, the natural reduction homomorphism

$$\mathrm{SL}_2(\mathbb{Z}) \rightarrow \mathrm{SL}_2(\mathbb{Z}/N\mathbb{Z})$$

is surjective with kernel $\Gamma(N)$. Therefore, we have an isomorphism

$$\mathrm{SL}_2(\mathbb{Z})/\Gamma(N) \rightarrow \mathrm{SL}_2(\mathbb{Z}/N\mathbb{Z}).$$

In particular, $\Gamma(N)$ and hence $\Gamma_1(N)$ and $\Gamma_0(N)$ are all of finite index in $\mathrm{SL}_2(\mathbb{Z})$.

Lemma 1.0.2. Let $\Gamma \subset \mathrm{SL}_2(\mathbb{Z})$ be a congruence subgroup of $\mathrm{SL}_2(\mathbb{Z})$ (Definition 1.0.1).

There is some integer $h \in \mathbb{Z}_{>0}$ such that $\begin{pmatrix} 1 & h \\ 0 & 1 \end{pmatrix} \in \Gamma$.

Definition 1.0.3 (Modular group action). Let $\Gamma \subseteq \mathrm{SL}_2(\mathbb{Z})$ be a subgroup. For $\gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \Gamma$ and $\tau \in \mathbb{H} := \{z \in \mathbb{C} \mid \mathrm{Im}(z) > 0\}$, define the action

$$\gamma \cdot \tau := \frac{a\tau + b}{c\tau + d}.$$

This is called the *modular action of Γ on the upper half-plane $\mathbb{H}$* .

Definition 1.0.4 (Weight- k slash operator). Let $k \in \mathbb{Z}$. For $\gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{SL}_2(\mathbb{Z})$ and any function $f : \mathbb{H} \rightarrow \mathbb{C}$, the *weight- k slash operator* (or simply the *weight- k operator*) is defined as

$$(f|_k\gamma)(\tau) := (c\tau + d)^{-k} f\left(\frac{a\tau + b}{c\tau + d}\right).$$

$f|_k\gamma$ is also often denoted by $f[\gamma]_k$.

Definition 1.0.5 (Weakly modular form). Let $k \in \mathbb{Z}$, and let $\Gamma \subseteq \mathrm{SL}_2(\mathbb{Z})$ be a subgroup. A meromorphic function $f : \mathbb{H} \rightarrow \mathbb{C}$ is called a *weakly modular form of weight k with respect to Γ* if

$$f|_k\gamma = f \quad \text{for all } \gamma \in \Gamma.$$

(Definition 1.0.4).

Lemma 1.0.6. Let $\Gamma \subset \mathrm{SL}_2(\mathbb{Z})$ be a congruence subgroup of $\mathrm{SL}_2(\mathbb{Z})$ (Definition 1.0.1). Let $h \in \mathbb{Z}_{>0}$ be such that $\begin{pmatrix} 1 & h \\ 0 & 1 \end{pmatrix} \in \Gamma$ by Lemma 1.0.2. Every weakly modular form f (Definition 1.0.5) of some weight $k \in \mathbb{Z}$ is $h\mathbb{Z}$ -periodic, i.e. $f(z+h) = f(z)$ for all $z \in \mathbb{H}$ (♠ TODO: cite the upper half plane).

In particular, there exists a function $g : D' \rightarrow \mathbb{C}$ where $D' \subset \mathbb{C}$ is the punctured unit disk such that $f(z) = g(q_h)$ where $q_h = e^{2\pi iz/h}$.

Definition 1.0.7. Let $\Gamma \subset \mathrm{SL}_2(\mathbb{Z})$ be a congruence subgroup of $\mathrm{SL}_2(\mathbb{Z})$ (Definition 1.0.1).

Let $h \in \mathbb{Z}_{>0}$ be minimal such that $\begin{pmatrix} 1 & h \\ 0 & 1 \end{pmatrix} \in \Gamma$ by Lemma 1.0.2.

Let $f : \mathbb{H} \rightarrow \mathbb{C}$ be a meromorphic function of the form $f(z) = g(q_h)$ for some meromorphic function $g : D' \rightarrow \mathbb{C}$ on the punctured unit disk D' where $q_h = e^{2\pi iz/h}$ (e.g. weakly modular forms (Definition 1.0.5) satisfy this condition, see Lemma 1.0.6). In particular, g has a Laurent expansion near $q_h = 0$ (which corresponds to $\mathrm{Im}(z) \rightarrow \infty$) (♠ TODO: cite this as a complex analytic fact) and hence f has a Fourier expansion:

$$f(z) = \sum_{n=-\infty}^{\infty} a_n q_h^n.$$

The values $a_n \in \mathbb{C}$ are called the **Fourier coefficients of f** .

We say that f is **meromorphic at ∞** if g is meromorphic at $q_h = 0$, i.e. the Fourier expansion of f has the form

$$f(z) = \sum_{n=n_0}^{\infty} a_n q_h^n.$$

for some $n_0 \in \mathbb{Z}$. We say that f is **holomorphic at ∞** if g is holomorphic at $q_h = 0$, i.e. the Fourier expansion of f has the form

$$f(z) = \sum_{n=0}^{\infty} a_n q_h^n.$$

If f is holomorphic at ∞ and we additionally have $a_0 = 0$, then we say that f has a **cusp at ∞** .

Definition 1.0.8. (♠ TODO: define holomorphic function, upper half plane) (♠ TODO: define cusps of Γ) Let k be an integer and $\Gamma \leq \mathrm{SL}_2(\mathbb{Z})$ a congruence subgroup (Definition 1.0.1). A **modular form of weight k with respect to Γ** is a function $f : \mathbb{H} \rightarrow \mathbb{C}$, where $\mathbb{H}$ is the upper half-plane, such that

- The function f is holomorphic on $\mathbb{H}$.
- The function f is a weakly modular form of weight k with respect to Γ (Definition 1.0.5). Equivalently, for all $\gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \Gamma$ and all $z \in \mathbb{H}$, we have

$$f\left(\frac{az+b}{cz+d}\right) = (cz+d)^k f(z).$$

- f is holomorphic at the cusps of Γ (i.e., its Fourier expansion at all cusps of Γ is holomorphic). More precisely, $f|_k \alpha$ (Definition 1.0.4) is holomorphic at ∞ (Definition 1.0.7) for all $\alpha \in \Gamma$.

A **cusp form** is a modular form f whose Fourier expansion at every cusp of Γ has constant term 0, i.e. $f|_k \alpha$ has a cusp at ∞ for every $\alpha \in \Gamma$.

Definition 1.0.9. (♠ TODO: define holomorphic function, upper half plane) (♠ TODO: define cusps of Γ) Let k be an integer and $\Gamma \leq \mathrm{SL}_2(\mathbb{Z})$ a congruence subgroup (Definition 1.0.1). A **modular form of weight k with respect to Γ** is a function $f : \mathbb{H} \rightarrow \mathbb{C}$, where $\mathbb{H}$ is the upper half-plane, such that

- The function f is meromorphic on $\mathbb{H}$.

- The function f is a weakly modular form of weight k with respect to Γ (Definition 1.0.5). Equivalently, for all $\gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \Gamma$ and all $z \in \mathbb{H}$, we have

$$f\left(\frac{az+b}{cz+d}\right) = (cz+d)^k f(z).$$

- f is meromorphic at the cusps of Γ (i.e., its Fourier expansion at all cusps of Γ is holomorphic). More precisely, $f|_k \alpha$ (Definition 1.0.4) is meromorphic at ∞ (Definition 1.0.7) for all $\alpha \in \Gamma$.

We define the L -function of a modular form by putting the Fourier coefficients of the modular form into a Dirichlet series.

Definition 1.0.10 (L-function of a modular form). Let $k \in \mathbb{Z}_{>0}$, and let $f : \mathbb{H} \rightarrow \mathbb{C}$ be a modular form (Definition 1.0.8) of weight k with respect to a congruence subgroup (Definition 1.0.1) $\Gamma \subseteq \mathrm{SL}_2(\mathbb{Z})$. Let

$$f(\tau) = \sum_{n=0}^{\infty} a_n e^{\frac{2\pi i n \tau}{h}}$$

be the Fourier expansion of f (Definition 1.0.7) where $h > 0$ is an appropriate integer.

The *L-function of the modular form f* is the complex function defined for $s \in \mathbb{C}$ with sufficiently large $\mathrm{Re}(s)$ by the Dirichlet series

$$L(f, s) := \sum_{n=1}^{\infty} \frac{a_n}{n^s}.$$

(♠ TODO: define the completed L -function)

The construction of the L -function of a modular form is not arbitrary — a Fourier transform involving the modular form yields the completed L -function.

Theorem 1.0.11. Let $k \in \mathbb{Z}_{>0}$, and let $f : \mathbb{H} \rightarrow \mathbb{C}$ be a modular form (Definition 1.0.8) of weight k with respect to a congruence subgroup (Definition 1.0.1) $\Gamma \subseteq \mathrm{SL}_2(\mathbb{Z})$. Let

$$f(\tau) = \sum_{n=0}^{\infty} a_n e^{\frac{2\pi i n \tau}{h}}$$

be the Fourier expansion of f (Definition 1.0.7) where $h > 0$ is an appropriate integer.

The Mellin transform of $f(i\tau)$ equals the completed L -function $\Lambda(f, s)$.

Proof. Let $f : \mathbb{H} \rightarrow \mathbb{C}$ be a meromorphic function periodic with period $h > 0$, so

$$f(z+h) = f(z).$$

This implies that f factors through the variable

$$q_h := e^{2\pi i z/h}.$$

Define

$$\Phi(t) := f(it), \quad t > 0.$$

Using the periodicity, f has a Fourier expansion in q_h :

$$f(z) = \sum_{n=n_0}^{\infty} a_n q_h^n = \sum_{n=n_0}^{\infty} a_n e^{2\pi i n z/h}.$$

Substituting $z = it$, we have

$$\Phi(t) = f(it) = \sum_{n=n_0}^{\infty} a_n e^{-2\pi n t/h}.$$

Consider the Mellin transform integral with complex parameter s :

$$I(s) := \int_0^{\infty} \Phi(t) t^s \frac{dt}{t}.$$

For $\operatorname{Re}(s)$ sufficiently large, interchange sum and integral (justified by absolute convergence):

$$I(s) = \sum_{n=n_0}^{\infty} a_n \int_0^{\infty} e^{-2\pi n t/h} t^s \frac{dt}{t}.$$

Evaluate the integral inside for each fixed n . Change variables by setting $u = \frac{2\pi n t}{h}$, so $t = \frac{hu}{2\pi n}$, and

$$dt = \frac{h}{2\pi n} du.$$

Hence,

$$\int_0^{\infty} e^{-2\pi n t/h} t^s \frac{dt}{t} = \int_0^{\infty} e^{-u} \left(\frac{hu}{2\pi n} \right)^s \frac{du}{u} = \left(\frac{h}{2\pi n} \right)^s \int_0^{\infty} e^{-u} u^s \frac{du}{u}.$$

Recognizing the Gamma function, we get

$$\int_0^{\infty} e^{-u} u^s \frac{du}{u} = \Gamma(s).$$

Therefore,

$$I(s) = \Gamma(s) \left(\frac{h}{2\pi} \right)^s \sum_{n=n_0}^{\infty} a_n n^{-s}.$$

By definition, the completed L -function $\Lambda(f, s)$ associated to f is proportional to this Mellin transform expression:

$$\Lambda(f, s) = \Gamma(s) \left(\frac{h}{2\pi} \right)^s L(f, s),$$

where

$$L(f, s) := \sum_{n=n_0}^{\infty} \frac{a_n}{n^s}.$$

This establishes the integral representation of $\Lambda(f, s)$ as the Mellin transform of $\Phi(t) = f(it)$, adapted to period h . $\square$

1.1. L -function of a modular form.

2. AUTOMORPHIC FORMS

Definition 2.0.1 (Adelic Quotient). Let F be a global field (Definition B.0.1), and let $\mathbb{A}_F$ be the ring of adeles (Definition B.0.4) of F . Let G be a linear algebraic group (Definition B.0.7) defined over F . The *adelic quotient of G over F* is the quotient space

$$G(F) \backslash G(\mathbb{A}_F).$$

(Definition B.0.5) where $G(F)$ is given the discrete topology. (♠ TODO: What topology does $G(F)$ have?)

Definition 2.0.2 (Automorphy). Let F be a global field (Definition B.0.1), and let $\mathbb{A}_F$ be the ring of adeles of F (Definition B.0.4). Let G be a linear algebraic group (Definition B.0.7) defined over F .

Let $f : G(\mathbb{A}_F) \rightarrow \mathbb{C}$ be a function. We say f satisfies *automorphy* if it is left $G(F)$ -invariant, i.e.,

$$f(\gamma g) = f(g) \quad \text{for all } \gamma \in G(F), g \in G(\mathbb{A}_F).$$

Equivalently, f is a function which descends to a well defined function $f : G(F) \backslash G(\mathbb{A}_F) \rightarrow \mathbb{C}$ on the adelic quotient (Definition 2.0.1).

Definition 2.0.3 (Right-regular action of $G(\mathbb{A}_F)$). Let F be a global field (Definition B.0.1), and let $\mathbb{A}_F$ be the ring of adeles (Definition B.0.4) of F . Let G be a linear algebraic group (Definition B.0.7) defined over F . For $g \in G(\mathbb{A}_F)$ (Definition B.0.5) and a function $f : G(F) \backslash G(\mathbb{A}_F) \rightarrow \mathbb{C}$, the *right-regular action/representation of $G(\mathbb{A}_F)$* is the representation

$$R : G(\mathbb{A}_F) \rightarrow \text{Aut}(\text{Fun}(G(F) \backslash G(\mathbb{A}_F), \mathbb{C}))$$

on the space of complex valued functions on $G(F) \backslash G(\mathbb{A}_F)$ given by

$$(R(g)f)(x) = f(xg)$$

for $g \in G(\mathbb{A}_F)$, $f : G(F) \backslash G(\mathbb{A}_F) \rightarrow \mathbb{C}$, and $x \in G(F) \backslash G(\mathbb{A}_F)$.

Definition 2.0.4 (Smooth function). Let F be a global field (Definition B.0.1), and let $\mathbb{A}_F$ be the ring of adeles of F (Definition B.0.4). Let G be a linear algebraic group (Definition B.0.7) defined over F .

Let $f : G(F) \backslash G(\mathbb{A}_F) \rightarrow \mathbb{C}$ be a function. We say f is *smooth* if it is (♠ TODO: define $G(\mathbb{A}_F)$, $G(\mathbb{A}_F^{\text{fin}})$, $G(F_\infty)$)

1. locally constant in the non-archimedean directions, i.e. for every point $g = (g_v)_v \in G(\mathbb{A}_F)$, there exists an open compact subgroup $U = \prod_v U_v \subset G(\mathbb{A}_F^{\text{fin}})$ (Definition B.0.5) such that for all $u \in U$, we have

$$f(gu) = f(g),$$

and

2. C^∞ in each archimedean direction, i.e. for every point $g = (g_v)_v \in G(\mathbb{A}_F)$, the map

$$G(F_\infty) \rightarrow \mathbb{C}, (h_v)_{v|\infty} \mapsto f(g \cdot (h_v)_{v|\infty})$$

(Definition B.0.5) is C^∞ as a smooth manifold; note that $G(F_\infty) = \prod_{v|\infty} G(F_v)$ is a product of real and complex Lie groups. (♠ TODO: define real and complex lie groups)

Definition 2.0.5 (K -finiteness). Let F be a global field (Definition B.0.1), and let $\mathbb{A}_F$ be the ring of adeles of F (Definition B.0.4). Let G be a linear algebraic group (Definition B.0.7) defined over F .

Let $K \subseteq G(\mathbb{A}_F)$ be a maximal compact subgroup. A smooth function (Definition 2.0.4) $f : G(F) \backslash G(\mathbb{A}_F) \rightarrow \mathbb{C}$ is called **K -finite** if the vector space

$$\langle R(k)f : k \in K \rangle$$

(Definition 2.0.3) spanned by the K -translates of f under the right-regular action (Definition 2.0.3) is finite-dimensional.

Definition 2.0.6 (Moderate growth). Let F be a global field (Definition B.0.1), and let $\mathbb{A}_F$ be the ring of adeles of F (Definition B.0.4). Let G be a linear algebraic group (Definition B.0.7) defined over F . Let $\|\cdot\|$ be a fixed height function (or norm-like function) on $G(\mathbb{A}_F)$ compatible with F and G (e.g. the height function of Definition B.0.6).

A smooth function (Definition 2.0.4) $f : G(F) \backslash G(\mathbb{A}_F) \rightarrow \mathbb{C}$ is said to have **moderate growth** if there exist constants $C > 0$ and $N \geq 0$ such that

$$|f(g)| \leq C \cdot \|g\|^N \quad \text{for all } g \in G(\mathbb{A}_F),$$

Definition 2.0.7 (Automorphic form). Let F be a global field (Definition B.0.1), and let $\mathbb{A}_F$ be the ring of adeles of F (Definition B.0.4). Let G be a linear algebraic group (Definition B.0.7) defined over F .

A function $f : G(F) \backslash G(\mathbb{A}_F) \rightarrow \mathbb{C}$ is called an **automorphic form** if it satisfies all of the following conditions:

1. f is smooth (Definition 2.0.4),
2. f is K -finite (Definition 2.0.5) for some maximal compact subgroup $K \subseteq G(\mathbb{A}_F)$,
3. f has moderate growth (Definition 2.0.6),

(♠ TODO: read the following AI generated statements and make them precise) (♠ TODO: Formulate how modular forms are automorphic forms)

Theorem 2.0.8 (Basic Properties of Automorphic Forms). Let F be a global field, $\mathbb{A}_F$ its ring of adeles, and G a linear algebraic group defined over F . Let $K \subseteq G(\mathbb{A}_F)$ be a maximal compact subgroup. Suppose

$$f : G(F) \backslash G(\mathbb{A}_F) \rightarrow \mathbb{C}$$

is an automorphic form, i.e., f is smooth, K -finite, and of moderate growth. Then:

1. f is left-invariant under $G(F)$ (automorphy condition).

2. f generates a $(\mathfrak{g}, K)$ -module under the right regular representation of $G(\mathbb{A}_F)$, where $\mathfrak{g}$ is the Lie algebra of $G(F_\infty)$.
3. The space of automorphic forms is preserved by Hecke operators arising from elements in the spherical Hecke algebra associated with $G(\mathbb{A}_F^\infty)$.

Proposition 2.0.9 (Equivalent Characterizations of Automorphic Forms). Under standard assumptions on G , the following conditions on a function $f : G(F) \backslash G(\mathbb{A}_F) \rightarrow \mathbb{C}$ are equivalent:

1. f is smooth, K -finite, of moderate growth, and left $G(F)$ -invariant.
2. f corresponds to a vector in a smooth admissible representation of $G(\mathbb{A}_F)$ realized as a subrepresentation of the space of L^2 automorphic forms.

Corollary 2.0.10 (Automorphic Forms as Sections). Automorphic forms can be identified with smooth sections of certain automorphic vector bundles over the quotient space $G(F) \backslash G(\mathbb{A}_F) / K$ satisfying growth conditions.

Remark 2.0.11 (Intuition). Automorphic forms generalize classical modular forms by encoding arithmetic and representation-theoretic data through functions on adelic quotients. The conditions of smoothness, K -finiteness, and moderate growth control analytic and algebraic properties needed for harmonic analysis and spectral theory on these quotients.

APPENDIX A. MISCELLANEOUS DEFINITIONS

Definition A.0.1 (Topological groups). A *topological group* is a group $(G, \cdot)$ together with a topology $\mathcal{T}$ on G such that the maps

$$\begin{aligned} \mu : G \times G &\rightarrow G, & (g, h) &\mapsto g \cdot h, \\ \iota : G &\rightarrow G, & g &\mapsto g^{-1}, \end{aligned}$$

are continuous with respect to the product topology on $G \times G$ and the topology $\mathcal{T}$ on G .

A *abelian topological group* or a *commutative topological group* is a topological group that is abelian as a group.

APPENDIX B. ADÈLES AND IDÉLES OF GLOBAL FIELDS

Definition B.0.1. A *global field* is a field K that is either:

- a number field, i.e. a finite extension of $\mathbb{Q}$, or
- a global function field, i.e. a finite extension of the field of rational functions $\mathbb{F}_q(t)$ in one variable over a finite field $\mathbb{F}_q$.

Definition B.0.2. Let K be a global field (Definition B.0.1) and let v be a place of K . Write $|\cdot|_v$ for an absolute value representing v . The *completion of K at v* , often denoted K_v , is the completion of K with respect to the metric induced by $|\cdot|_v$.

Definition B.0.3. Let $\{X_i\}_{i \in I}$ be a family of topological spaces indexed by a set I . For each $i \in I$, let $K_i \subseteq X_i$ be a topological subspace.

The *restricted product topology* on the restricted product

$$\prod'_{i \in I} X_i := \left\{ (x_i)_{i \in I} \in \prod_{i \in I} X_i \mid x_i \in K_i \text{ for all but finitely many } i \in I \right\},$$

with respect to the subsets $\{K_i\}_{i \in I}$, is the coarsest topology such that:

- The natural inclusion maps $X_j \rightarrow \prod'_{i \in I} X_i$, defined by $x_j \mapsto (y_i)$ where $y_j = x_j$ and $y_i = k_i$ (a fixed element in K_i) for all $i \neq j$, are continuous for all $j \in I$.
- The subspace topology on the product $\prod_{i \in F} X_i$ for any finite subset $F \subseteq I$ (where coordinates outside F are fixed in K_i) coincides with the product topology on finitely many factors.

Equivalently, the restricted product topology is generated by the base consisting of sets of the form

$$\prod_{i \in F} U_i \times \prod_{i \notin F} K_i,$$

where $F \subseteq I$ is finite, U_i are open sets in X_i , and outside F the coordinates lie in K_i .

Definition B.0.4. Let K be a global field (Definition B.0.1). Write M_K for the set of all places of K and write M_K^∞ for the set of archimedean places of K . Let $S \subseteq M_K$ be some subset of places of K (typically, S is a finite set). For each $v \in M_K$, write $\mathcal{O}_v$ for the ring of integers in the completion K_v (Definition B.0.2)

- The *adèle ring of K* , denoted $\mathbb{A}_K$, is the restricted direct product (Definition B.0.3) of the K_v (over all places v of K), with respect to the $\mathcal{O}_v$ at non-archimedean v :

$$\mathbb{A}_K = \left\{ (x_v)_v \in \prod_{v \in M_K} K_v \mid x_v \in \mathcal{O}_v \text{ for all but finitely many non-archimedean } v \right\}.$$

- The *idèle group of K* , commonly denoted $\mathbb{A}_K^\times$ or $\mathbb{I}_K$, is the group of invertible elements of $\mathbb{A}_K$:

$$\mathbb{I}_K = \mathbb{A}_K^\times = \left\{ (x_v)_v \in \prod_{v \in M_K} K_v^\times \mid x_v \in \mathcal{O}_v^\times \text{ for all but finitely many non-archimedean } v \right\},$$

where $\mathcal{O}_v^\times$ denotes the group of units of $\mathcal{O}_v$ for non-archimedean v .

- The *adèle ring outside S of K* , commonly denoted $\mathbb{A}_K^S$ or $\mathbb{A}_{K,S}$, is the restricted product of the completions K_v over all places $v \in M_K \setminus S$, with respect to the rings of integers $\mathcal{O}_v$ at non-archimedean places:

$$\mathbb{A}_{K,S} = \mathbb{A}_K^S = \left\{ (x_v)_v \in \prod_{v \in M_K \setminus S} K_v \mid x_v \in \mathcal{O}_v \text{ for all but finitely many non-archimedean } v \right\}.$$

- The *idèle group outside S of K* , commonly denoted $(\mathbb{A}_K^\times)^S$, $(\mathbb{A}_{K,S}^\times)$, $\mathbb{I}_K^S$, or $\mathbb{I}_{K,S}$ is the group of invertible elements of $\mathbb{A}_K^S$:

$$(\mathbb{A}_K^\times)^S = \left\{ (x_v)_v \in \prod_{v \in M_K \setminus S} K_v^\times \mid x_v \in \mathcal{O}_v^\times \text{ for all but finitely many non-archimedean } v \right\}.$$

- The *ring of finite adèles of K* , commonly denoted $\mathbb{A}_{K,\text{fin}}$, $\mathbb{A}_K^{\text{fin}}$, $\mathbb{A}_{K,f}$, $\mathbb{A}_K^f$, is the adèle ring outside $S = M_K^\infty$, the set of archimedean places of K :

$$\mathbb{A}_{K,\text{fin}} := \mathbb{A}_K^{M_K^\infty} = \left\{ (x_v)_v \in \prod_{v \notin M_K^\infty} K_v \mid x_v \in \mathcal{O}_v \text{ for all but finitely many non-archimedean } v \right\}.$$

- The *finite idèle group of K* , commonly denoted $\mathbb{A}_{K,\text{fin}}^\times$, $\mathbb{I}_{K,\text{fin}}$, $\mathbb{I}_K^{\text{fin}}$, $\mathbb{I}_{K,f}$, $\mathbb{I}_K^f$ etc. is the group of units of the ring of finite adèles:

$$\mathbb{A}_{K,\text{fin}}^\times := (\mathbb{A}_K^\times)^{M_K^\infty} = \left\{ (x_v)_v \in \prod_{v \notin M_K^\infty} K_v^\times \mid x_v \in \mathcal{O}_v^\times \text{ for all but finitely many non-archimedean } v \right\}.$$

All of these are equipped with the restricted product topology (Definition B.0.3) induced by the topologies of the local fields K_v and the subspace topologies thereof.

Definition B.0.5 (Adelic points of an algebraic group). Let K be a global field (Definition B.0.1) with ring of integers $\mathcal{O}_K$, and let G be a linear algebraic group (Definition B.0.7) defined over K .

- The *group of adelic points of G* is the group $G(\mathbb{A}_K)$ defined by

$$G(\mathbb{A}_K) := \prod'_v G(K_v),$$

(Definition B.0.2) where the restricted product is taken with respect to the compact open subgroups $G(\mathcal{O}_v)$ for almost all non-archimedean v .

- For a set S (usually finite) of places of K , the *group of adelic points of G outside S* is the group $G(\mathbb{A}_{K,S})$ defined by

$$G(\mathbb{A}_{K,S}) := \prod'_{v \notin S} G(K_v).$$

- The *group of finite adelic points* is the group $G(\mathbb{A}_{K,\text{fin}}) = G(\mathbb{A}_K^{\text{fin}})$ defined by

$$G(\mathbb{A}_K^{\text{fin}}) := \prod'_{v \text{ non-archimedean}} G(K_v).$$

- The *group of archimedean adelic points* is the group $G(K_\infty)$ defined by

$$G(K_\infty) := \prod_{v \text{ archimedean}} G(K_v).$$

Thus one has a natural factorization

$$G(\mathbb{A}_K) \cong G(\mathbb{A}_K^{\text{fin}}) \times G(K_\infty).$$

The groups $G(\mathbb{A}_K)$, $G(\mathbb{A}_{K,S})$, and $G(\mathbb{A}_K^{\text{fin}})$ are all topological groups (Definition A.0.1) under the restricted product topology (Definition B.0.3) induced by the topologies of the local fields K_v and the subspace topologies of $\mathcal{O}_v$ and become topological groups. The group $G(K_\infty)$ is a topological group under the direct product topology.

Definition B.0.6 (Height function on $G(\mathbb{A}_F)$). Let F be a global field (Definition B.0.1), and let $\mathbb{A}_F$ be the ring of adeles of F (Definition B.0.4). Let G be a linear algebraic group (Definition B.0.7) defined over F embedded as an algebraic subgroup in GL_n .

For each place v of F , let $\|\cdot\|_v$ be a norm on $G(F_v) \subseteq \text{GL}_n(F_v)$ defined as follows:

$$\|g_v\|_v := \max_{1 \leq i, j \leq n} |(g_v)_{ij}|_v,$$

(♠ TODO: define the usual absolute value on F_v) where $|\cdot|_v$ is the usual absolute value or valuation on F_v .

Then, for $g = (g_v)_v \in G(\mathbb{A}_F)$, define the adelic height function by the product

$$\|g\| := \prod_v \|g_v\|_v,$$

where the product converges due to $\|g_v\|_v = 1$ for all but finitely many non-archimedean places.

Definition B.0.7 (Linear algebraic group over a scheme). Let S be a base scheme. A *linear algebraic group over S* is an affine group scheme G over S that is finitely presented and smooth over S , and such that for some integer $n \geq 1$, there exists a closed immersion of S -group schemes

$$G \hookrightarrow \text{GL}_{n,S}.$$

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